Journal of Experimental Marine Biology and Ecology 481 (2016) 15–24

Contents lists available at ScienceDirect

Journal of Experimental Marine Biology and Ecology

journal homepage: www.elsevier.com/locate/jembe

Patterns of surface lesion recovery in the Northern Star Coral, *Astrangia poculata*

Lukas DeFilippo [a,b,1], Elizabeth M. Burmester [a,b], Les Kaufman [a,b], Randi D. Rotjan [a,*]

[a] John H. Prescott Marine Laboratory, New England Aquarium, 1 Central Wharf, Boston, MA 02110, USA
[b] Boston University Department of Biology, 5 Cummington Mall, Boston, MA 02115, USA

ARTICLE INFO

Article history:
Received 22 October 2015
Received in revised form 22 March 2016
Accepted 23 March 2016
Available online 30 April 2016

Keywords:
Astrangia poculata
Facultative symbiosis
'Phoenix effect'
Lesion recovery
Temperate coral

ABSTRACT

Corals regularly experience partial mortality due to grazing, sedimentation, disease, and abrasion. Because a colony's ability to heal such surface lesions can reflect its overall health, recovery from artificial wounds has been posed as a potential in situ measure of coral status and resilience. In this study, the response of a temperate, facultatively symbiotic coral, *Astrangia poculata*, to wounds of varying size and shape was measured, along with the influence of *Symbiodinium psygmophilum*. Because most tropical corals host an obligate symbiosis, the impact of the symbionts on lesion recovery cannot be quantified separately from the hosts. Here, symbiotic and aposymbiotic *A. poculata* colonies were subjected to lesions representing 25%, 50%, and 75% total colony loss, with varying wound shape. Colonies were monitored for recovery, mortality of living tissue, photosynthetic efficiency and approximated chlorophyll density. Recovery was observed to occur primarily from within the lesion site, demonstrating a 'Phoenix effect' healing pattern whereby recovery stems from tissue remnants in the wound area. Recovery was best explained by an interaction effect between symbiotic state (symbiotic versus aposymbiotic) and the amount of residual tissue visible after wounding. Per unit area of residual tissue, healing was greater in symbiotic colonies. Symbiotic colonies also exhibited higher estimates of photochemical efficiency and approximated chlorophyll density, likely contributing to their enhanced healing capacity. These results support previously documented reductions in wound recovery among bleached corals, and suggest that symbiont loss itself, rather than associated stress or physiological disruption contributes to the diminished healing capacity of bleached colonies. Tissue mortality in un-wounded colony portions was greatest among symbiotic corals exposed to 50% colony loss amassed over two wounds, suggesting that corals with multiple injuries may experience greater tissue loss compared to single-wounded colonies, which can help inform methods for wound proxy metrics in the field. Dissected decalcified *A. poculata* colonies revealed deep (~4–6 mm) corallite and polyp penetration, suggesting that surface lesions in this species are likely to leave tissue remnants buried within the skeleton. As such, it appears that *A. poculata* and corals with similar morphologies are predisposed to 'Phoenix effect' recovery, since most natural lesions would not fully penetrate the full depth of their corallites.

© 2016 Elsevier B.V. All rights reserved.

1. Introduction[2]

Partial mortality is a constant source of sublethal stress for most colonial, sessile, and modular organisms (Mistri and Ceccherelli, 1995; Harmelin and Marinopoulos, 1994; Meszaros and Bigger, 1999; Jackson and Winston, 1981; Crawley, 1983). The ability of these organisms to repair wounds quickly and successfully can have major consequences for their survival, growth, and reproduction (Meesters et al., 1996; Rinkevich, 1996; Rinkevich and Loya, 1989; Meesters et al., 1994; Rotjan and Lewis, 2005). Because of the ecological and adaptive importance of wound healing, conservation and monitoring efforts are beginning to explore recovery from artificial wounds as a proxy for organismal status and resilience to a variety of stressors (as in Fisher et al., 2007; Meesters and Bak, 1994; Meesters et al., 1992). However, wounds are highly variable (Meesters et al., 1997b; Lirman, 2000) and relationships between wound characteristics and recovery must be established before lesion healing can serve as a reliable proxy for organismal status.

Partly due to its utility as a proxy for colony health and performance, wound healing has been relatively well studied in corals. One of the earliest hypotheses generated by this research was 'localized regeneration' (LR), which posits that the energy for recovery is provided entirely by a narrow region of living tissue directly bordering the lesion (Meesters et al., 1994, 1997a; Bak and Steward-Van Es, 1980). Consequently, this hypothesis purports that the ratio of a wound's perimeter to its surface

* Corresponding author.
E-mail address: rrotjan@neaq.org (R.D. Rotjan).
[1] Present address: School of Aquatic and Fishery Sciences, University of Washington, 1122 NE Boat Street, Seattle, WA 98105, USA.
[2] RGB: Red, Green, Blue.
P:SA: Perimeter to surface area ratio.

http://dx.doi.org/10.1016/j.jembe.2016.03.016
0022-0981/© 2016 Elsevier B.V. All rights reserved.

area (P:SA) is the primary determinant of healing capacity (Meesters et al., 1997a; Van Woesik, 1998). Studies spanning multiple species support this hypothesis by demonstrating a positive functional relationship between healing rate and lesion P:SA (Meesters et al., 1997a; Van Woesik, 1998). Under LR, colony size has no influence on healing capacity, although small colonies may be unable to withstand extensive lesioning and suffer whole colony mortality (Meesters et al., 1996).

While the LR hypothesis is well supported in the literature, there are alternative healing patterns documented in corals. Oren et al. (1997, 2001) described a recovery model involving more widespread 'colony integration' (CI). These authors observed only relatively small lesions (<1 cm^2) behaving as expected under the LR hypothesis, with larger wounds drawing resources from portions of the colony as distant as 10–15 cm from the lesion site. Thus, for larger lesions, recovery did not depend solely on a thin band of adjacent tissue, but on the colony's total size and available energy reserves. Importantly, both CI and LR describe recovery that originates from the lateral margins of the wound, driven by contributions from the living colony (Meesters et al., 1997a; Oren et al., 1997, 2001).

In addition to the LR and CI models, there is a third, functionally distinct recovery pattern documented in corals. Krupp et al. (1992) observed recovery from osmotically induced mortality in *Fungia scutaria* to be driven by cryptic residual tissue within the skeleton. Despite complete withdrawal of surface tissues, the authors found intact messentarial filaments and undifferentiated tissue 2–7 mm beneath the skeletal surface. This recovery pattern, creatively dubbed the 'Phoenix effect' (PE), was also implicated in the rapid recovery of *Porites* colonies after a mass bleaching event (Roff et al., 2014). Jokiel et al. (1993) also described recovery of *Porites* colonies after a freshwater kill to be driven by undifferentiated tissue within the coral skeleton, and Meesters et al. (1992) noted accelerated recovery of shallow lesions in *Porites asteroides* driven by tissue remnants in the skeleton. To date, most published descriptions of PE recovery have focused on ecosystem-wide patterns, and in depth descriptions at the organism level have yet to be reported.

Like most corals, the taxa in which wound healing has been previously researched, (*Acropora*, *Montipora*, *Diploria*, *Porites*, *Agaricia*, *Orbicella*, *Favia*, *Siderastrea*, and *Fungia*) (Hall, 2001; Meesters et al., 1996; Titlyanov et al., 2008; Denis et al., 2011; Jayewardene, 2010; Van Woesik, 1998; Bak and Steward-Van Es, 1980; Meesters et al., 1997a,b; Fisher et al., 2007; Mascarelli and Bunkley-Williams, 1999; Oren et al., 1997, 2001; Kramarsky-Winter and Loya, 2000; Kramarsky-Winter and Loya, 1996) all maintain obligate symbioses with photosynthetic dinoflagellates (*Symbiodinium* spp.). Interruption of this symbiosis, commonly known as 'bleaching', is usually accompanied by stress and physiological disruption (Glynn et al., 1985), confounding comparisons between bleached and healthy corals for assessing the role of *Symbiodinium* in wound recovery. While previous studies have been crucial to elucidating the general patterns and mechanisms of coral lesion healing, these efforts did not determine the role of algal symbiosis in recovery. As has been previously noted, determining the benefits of symbiosis to the host and symbionts under normal conditions has lagged behind understanding symbiosis during stress and bleaching (Edmunds and Gates, 2003).

The Northern Star Coral, *Astrangia poculata* (=*Astrangia danae*; Peters et al., 1988), is a temperate coral that facultatively associates with *Symbiodinium psygmophilum* (LaJeunesse et al., 2012; Dimond and Carrington, 2007, 2008), sympatrically displaying multiple visually distinct 'symbiotic states' (sensu Dimond and Carrington, 2007; Mitchelmore et al., 2003) in shallow waters: brown, symbiotic, and white, aposymbiotic, with some intermediate colonies. The aposymbiotic state retains chronically reduced symbiont densities, maintained by proportionally greater expulsion rates than their fully symbiotic counterparts (Dimond and Carrington, 2008). As such, while aposymbiotic colonies are not completely azooxanthellate, they do not functionally receive any photosynthetic benefits from *S. psygmophilum*. This unique intraspecific dichotomy allows examination of the impact of *Symbiodinium* on various aspects of coral biology, without the confounding effects of stress and physiological disruption associated with bleaching in obligate symbiotic taxa (Glynn et al., 1985). The Northern Star Coral has already been used to investigate the impacts of algal symbiosis on host nutrition (Szmant-Froelich and Pilson, 1980, 1984), resistance to acidification (Holcomb et al., 2010, 2012), vulnerability to sedimentation (Peters and Pilson, 1985), calcification, and metabolism (Jacques and Pilson, 1980; Jacques et al., 1983; Cummings, 1983), and has proved to be a tractable experimental system with exciting potential for addressing fundamental questions of coral biology.

This study aimed to 1) directly evaluate the influence of *Symbiodinium* symbiosis on recovery and maintenance in *A. poculata*, and 2) describe the physical and temporal patterns of wound recovery in *A. poculata*, and the extent to which it adheres to any of the previously described healing models (localized regeneration (LR), colony integration (CI), or Phoenix effect (PE)). By evaluating the relationship between lesion P:SA, and colony size with healing capacity, as well as measuring any recovery from residual tissue fragments in the lesion site, it is possible to identify the dominant healing pattern(s) exhibited by *A. poculata*. Additionally, comparing recovery between symbiotic and aposymbiotic *A. poculata* provides novel insights into the contributions of *Symbiodinium* to coral wound healing and resilience.

2. Methods

2.1. Collection and husbandry

Coral specimens were collected in October 2013 from depths of 6–10 m at Fort Wetherill National Park in Jamestown, Rhode Island (41 28° 40°° N, 71 21° 34°° W) using SCUBA. Once collected, corals were acclimated for two months in an integrated, flow-through aquarium system supplied with ultraviolet-sterilized, particulate-filtered seawater from Boston Harbor. During acclimation, research specimens were removed of all visible epibionts, which included sponges, polychaete worms, and algae. After acclimation, colonies were randomly and evenly allocated among two integrated 30 gallon tanks such that approximately equal numbers from each symbiotic state and wound treatment group were present in each. Lighting was provided by high output (HO) T5 fluorescent fixtures (Hamilton Technology, Gardena, CA, Aruba Sun T5-V series) each housing one 54-watt daylight and actinic bulb. A stable temperature of 18 °C was maintained for the duration of the acclimation and experimental periods by a water chiller. Other water quality parameters, such as pH, nitrate, phosphate, calcium, and alkalinity were monitored and recorded weekly for the duration of the experiment. Coral specimens were placed on raised plastic grids, each containing sixteen to eighteen colonies in separate slots to limit inter-colonial antagonism. Each day, *A. poculata* specimens received targeted ad libitum feedings of copepods (JEHM Co., Inc.), and each aquarium was scrubbed and siphoned to remove uneaten food, detritus and algae. Individual colonies were identified with numbered tags that were affixed to the corals' non-living base with cyanoacrylate.

2.2. Experimental design

This experiment included eight distinct treatments (four wounding patterns among two symbiotic states) intended to directly model the effects of zooxanthellae symbiosis and lesion P:SA on recovery in *A. poculata* (Fig. 1). Coral specimens ($n = 50$ colonies, $n = 5$ of each symbiotic state per wound treatment and control group) were visually categorized as symbiotic or aposymbiotic, and verified as such using a chlorophyll proxy equation generated for *A. poculata* by Dimond and Carrington (2007). Colonies displaying polyps of mixed symbiotic state were not selected for the study, but changes in symbiotic state among individual polyps of aposymbiotic colonies were observed during the acclimation and experimental periods.

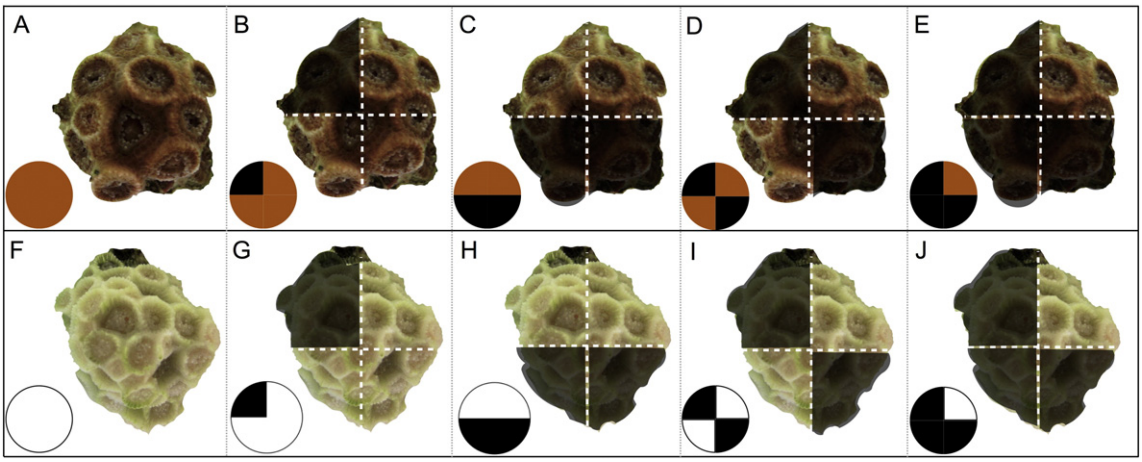

Fig. 1. Experimental design for colony wounding. Intact colonies (A, F), 25% (B, G), 50% adjacent (C, H), 50% separate (D, I), and 75% wounded. Brown and white circles represent wounding schema in symbiotic and aposymbiotic colonies of *A. poculata*, respectively. Shaded areas indicate wounded portions of the colony. N = 5 colonies for each wound group per symbiotic state.

For the designation of wounding pattern, the surface of each *A. poculata* colony was measured and considered as four quadrants of equal surface area. In quarter colony (25%) lesions, a single quadrant of the coral was ablated (Fig. 1 B, G). Two quadrants were ablated in the 50% adjacent (Fig. 1 C, H) and 50% separate (Fig. 1 D, I) lesion groups, adjacent to, and opposite of one another respectively to manipulate margin damage while keeping wound area constant. Three contiguous quadrants of the colony were wounded in the 75% lesion group (Fig. 1 E, J). These four wound types were intended to differ relatively predictably in their P:SA, with the greatest ratio in the 25% and 50% separate groups, followed by the 50% adjacent group, and lowest in the 75% wound group. Total wound perimeter estimates were also measured in ImageJ (NIH) so wound P:SA could be calculated for consideration as a continuous variable. Colony size was evaluated as a continuous variable based on the initial live surface area of each colony. Additionally, 10 colonies were allocated as controls (Fig. 1 A, F) and received no wounding treatment, although the photochemical efficiency of these corals was monitored during the study (Fig. 5 C).

2.3. Data collection

Prior to wounding, each *A. poculata* specimen was photographed from the top and all four sides to fully image the entire colony. A two-way ANOVA was used to assess potential biases in colony size (measured as live surface area) between symbiotic states and wound treatment groups. Initial live surface area of colonies was estimated by enveloping their living portions in aluminum foil, trimming it to account for creases and dead areas. The tin foil was then weighed, and a predetermined surface area to weight ratio for the foil was used to calculate the coral's surface area (Marsh, 1970). This method was selected over surface area measurements derived from whole-colony photographs because image analysis can only estimate surface area in two dimensions. For measuring recovery or mortality, this bias was less of an issue because the three-dimensionality of the colony is less pronounced over the localized spatial scales at which recovery/mortality was observed. However, for estimating an entire colony's size and designating wound area, planar estimates would have grossly underestimated the true surface area of the colony. Aposymbiotic corals exhibited significantly greater live surface area (11.78 ± 5.58 cm^2) compared to symbiotic colonies (8.48 ± 2.54 cm^2) ($F(1,44) = 7.548$, $p = 0.009$). However, this difference is reflective of the size distribution observed at the collection site (DeFilippo, personal observation). Colonies were randomly assigned to wound treatments, and no significant size bias was detected between wound groups ($F(4,44) = 1.508$ $p = 0.216$).

Once total live surface area had been measured, wound area was designated by assigning a percentage of the tin foil's mass corresponding to the coral's wound treatment group (e.g. 25, 50, 75%) for removal. All wound designations were made within $\pm 5\%$ of the target tin foil weight. Wounding consisted of first removing the visible polyp and coenenchyme tissue with a Waterpik©, and then eroding the raised corallite structure until relatively flat with a diamond coated file. No effort was made to destroy the remaining septa once the corallite structure had been leveled, nor to remove any tissue fragments visible in the lesion after the prescribed treatment. These wounds were not intended to replicate any specific natural lesioning agent, but constitute a generalized surface ablation such as might be incurred from a variety of mechanical stressors including storms and scraping corallivores (Woodley et al., 1981; Bonaldo et al., 2011).

Wounded *A. poculata* specimens were re-photographed within 48–72 h of wounding, and again at 7, 10, 15, 20, 30 and 60 days after wounding. These time points were selected to yield a fine-scale time series of the early stages of recovery, as well as coarser patterns occurring over longer time periods. Photographs were analyzed in ImageJ (NIH) to measure the surface area of recovered tissue and mortality of live tissue. Total estimates were made by summing planar measurements of these parameters across photographs from the top and all four sides of the coral, excluding redundant area across angles. Photographs for two coral specimens were deemed unreliable due to shifts in coral position across photographs, and were excluded from analysis. When measuring recovery, a distinction was made between tissue regenerating from the perimeter (characteristic of both LR and CI healing) versus the interior (central/PE recovery) of the lesion (Fig. 2).

Maximum quantum yield (F_v/F_m) measures the photosynthetic capacity of photosystem II (PS II), also known as photosynthetic efficiency. A Walz Junior pulse-amplitude modulated (PAM) fluorescence meter was used to measure the photosynthetic efficiency of each *A. poculata* colony prior to wounding, and again 3, 15, 30, and 60 days after wounding. Readings were taken between 1300 and 1500 h on each day to minimize the effects of diurnal fluctuations (Brown et al., 1999) on the time series data. After 30 min of dark acclimation in a close-topped black plastic container, individual colonies were transferred to a glass beaker filled with 2–3 in. of aquarium seawater that was placed within a dark plastic box where PAM readings were taken. A colony's photosynthetic efficiency at each time point was determined as the average maximum quantum yield measured from three haphazardly selected polyps. Briefly, 6 s of far red illumination was given to determine minimal fluorescence in the dark adapted state (F_0), and a saturating pulse (10,000 μmol m^{-2} s^{-1}) was applied over 0.6 s to determine maximum fluorescence (F_m). Illumination was applied by a light

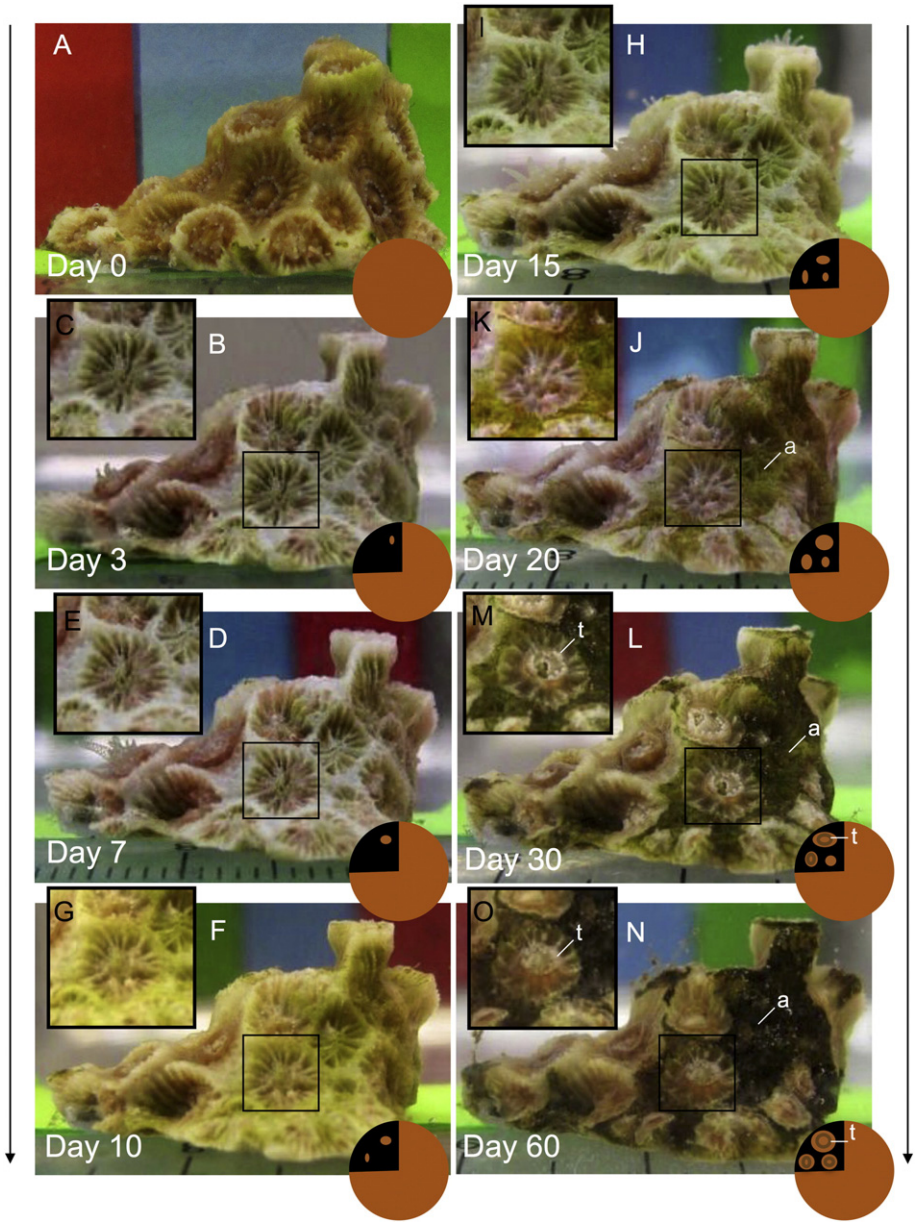

Fig. 2. Time series of wounding and recovery in *A. poculata*. Pre-wounding (A), after 3 days (B), 7 days (D), 10 days (F), 15 days (H), 20 days (J), 30 days (L), and 60 days (N). Insets: close-up of single polyp recovery (indicated with square) post-wounding after 3 days (C), 7 days (E), 10 days (G), 15 days (I), 20 days (K), 30 days (M), and 60 days (O). Newly formed tentacles denoted by "t," areas of algal overgrowth denoted by "a." Circles in lower right corner represent simplified diagrams of recovery: areas in brown signify live tissue cover while black regions signify removed tissue.

fiber held ~1 mm from the oral disk of each polyp at an angle of approximately 45–60°. The change in fluorescence (ΔF) caused by the saturating pulse was divided by maximal fluorescence to yield the maximum quantum yield, F_v/F_m.

Polyp color was analyzed from photographs taken before wounding, and 3, 15, 30, and 60 days after wounding. Polyp color values were obtained by quantifying red, green, and blue (RGB) pixels in 10 randomly selected polyps per colony against a color standard using a custom analysis script written in MATLAB (version R2007b, The MathWorks, Natick, MA). This method corrects for changes in absolute illumination and any color imparted by ambient light to allow comparison between images. Following the protocol of Dimond and Carrington (2007), RGB values were condensed to a single value using principal component analysis. The resulting principal component (PC) explained an average of 70.6 ± 4.72% of the variation in RGB, and component loadings averaged 88.4 ± 4.04, 93 ± 3.32, and 66 ± 18.53 for red, green, and blue respectively across all days measured. Principal component values were multiplied by −1 to induce realistic directionality, and were normalized to zero by adding the absolute value of the lowest transformed measurement to all values. To convert polyp color to approximated chlorophyll density, the adjusted PC values were applied to the equation developed by Dimond and Carrington (2007)

$$y = 0.044x^2 + 0.0335x \quad (1)$$

where y = approximated chlorophyll density (μg cm^{-2}) and x = polyp color PC ($R^2 = 0.89$). Although chlorophyll density is not a strict measurement of symbiont density because zooxanthellae pigment concentrations fluctuate (Warner et al., 2002; Cummings, 1983; Fitt et al., 2000), it has been shown to be a reliable symbiosis parameter for distinguishing symbiotic states in *A. poculata* (Dimond and Carrington, 2007). Additionally, in situ zooxanthellae counts require destructive tissue subsampling, which would inflict a second source of partial mortality to wounded colonies. The use of a symbiosis proxy (approximated

chlorophyll density) in *A. poculata* represents an important tool for enabling non-destructive assessments that do not interfere with or confound the recovery potential of colonies.

Additional wounded and control *A. poculata* specimens not associated with the wounding experiment were fixed in formaldehyde, and their skeletons were dissolved in hydrochloric acid for morphological observations of corallite and tissue penetration depth, allowing a more detailed examination via tissue dissection of the wounding treatment's effect on the lesion site.

2.4. Statistical analysis

All statistical analyses were performed using the R software package, version 3.2.0 (R Core Team (2015). All group means are reported as ± standard deviation. Because centrally recovered tissue indicates 'Phoenix effect' (PE) healing, and perimeter recovery is characteristic of both 'colony integration' (CI) and 'localized regeneration' (LR), the amount of central versus perimeter recovery observed was compared with Welch's paired t-test. For perimeter recovery, CI and LR can be further distinguished by a positive relationship between wound P:SA and colony size with recovery, respectively. Thus, wound P:SA, wound treatment, and colony size, in addition to symbiotic state and residual tissue (tissue visible in the lesion site within 48–72 h after wounding) were evaluated as potential independent variables when modeling recovery. Surface area of recovered tissue was divided by initial wound size and the resulting proportions were fitted to a beta regression model with a logit link function using the 'betareg' package, which accommodates proportional data bounded between 0 and 1 (Ferrari and Cribari-Neto, 2004).

Live tissue mortality was examined for potential associations with continuous independent variables such as colony surface area, but none were detected. Therefore, a factorial approach was used for analysis. A two-way ANOVA was used to compare relative tissue mortality across symbiotic states and wound treatment groups. The mortality data exhibited pronounced heteroscedasticity, which transformations failed to resolve. Since transformations did not correct the violation but altered the ANOVA outcomes, the untransformed results are presented, but should be interpreted cautiously.

To confirm the difference in symbiosis levels that distinguish symbiotic from aposymbiotic *A. poculata* colonies, approximated chlorophyll density was fitted to a restricted maximum likelihood (REML) linear mixed effects (LME) model, with symbiotic state, wound treatment group, and day as potential fixed effects, and coral specimen ID as a random effect. To compare functional photosynthetic efficiency between symbiotic states and assess possible variation among wound treatment groups, F_v/F_m values were also fitted to an LME model with the same suite of candidate fixed and random effects. While there was no biological basis to expect differences in photosynthetic efficiency or approximated chlorophyll density among wound treatment groups, it was important to assess whether such variation was present and potentially confounding the effects of the wounding treatments themselves on healing or mortality.

Model selection for wound recovery, F_v/F_m, and approximated chlorophyll density was based on Akaike's Information Criterion adjusted for small sample size (AIC_c), which balances model fit against overparameterization. Models containing every possible combination of candidate independent variables were fitted and ranked according to AIC_c, using the 'dredge' function of the 'MuMIn' package, and the difference between each model and the best model was calculated according to

$$\Delta_i = AIC_{c\,i} - AIC_{c\,min}. \quad (2)$$

A Δ_i of 0–2 indicates "substantial" support for model 'i', while 4–7 is "considerably less", and >10 is "essentially none" (Burnham and Anderson, 2002). Simultaneous Tukey pairwise comparisons were performed between levels of factors and interactions when necessary using the 'glht' function of the 'multcomp' package.

For all analyses, model assumptions were graphically assessed when necessary using quantile–quantile plots, density plots, and/or residual by predicted plots. For all response variables except mortality, wound treatment had no observed statistical effect. Therefore, all samples within each symbiotic state were pooled together for analyses, increasing the effective sample size to 20 colonies per symbiotic state.

3. Results

To determine the occurrence of central (PE) healing versus perimeter recovery (characteristic of both LR and CI recovery), the surface area of tissue originating within the lesion site from residual tissue fragments was compared to tissue regenerating from the margins of the wound. Perimeter recovery was observed in 15 out of the 38 (39.47%) recovering corals, accounting for only 5.77% of the total surface area of tissue recovered in this study. Among the colonies that did exhibit it, perimeter recovery constituted a relatively small portion of total recovery, averaging only 0.05 ± 0.05 cm^2, or 21.64 ± 21.01% (Fig. 3 D). By contrast, central recovery was observed in 100% of coral specimens examined, contributing an average of 0.35 ± 0.31 cm^2 of new tissue per coral, and accounting for 94.23% of total observed recovery (Fig. 3 D). Welch's paired t test confirmed that central recovery was the dominant healing pattern exhibited in this study ($t = 6.473, 38.248$, $p < 0.001$) (Fig. 3 D). The infrequency with which perimeter recovery was observed made explanatory statistical analyses intractable, so the final healing model describes only central (PE) recovery.

Recovery originated primarily from minor tissue remnants visible in the lesion site, which expanded and coalesced to eventually form differentiated structures (e.g. tentacles) and complete functional polyps. In all groups, recovered tissue increased sharply after wounding, stabilizing midway through the experimental period and eventually declining as undifferentiated tissue was overgrown by algae (Figs. 3 A–C, 2 J–N). Out of wound treatment, wound P:SA, colony surface area, symbiotic state, and residual tissue, the most parsimonious beta regression model of recovery included only symbiotic state and residual tissue, with an interaction between the two effects (Table 1). The most plausible models that did include initial live surface area, P:SA, or wound treatment lacked substantial support (Table 2). Additionally, each of these models was simply the best model (containing only the interaction between residual tissue and symbiotic state), with either initial live surface area, P:SA, or wound treatment added (Table 2). That these models showed reduced plausibility despite added complexity suggests that these additional parameters were not informative, and that the best model was that which contained only the interaction between residual tissue and symbiotic state. Final recovery was similar among aposymbiotic (20.04 ± 13.47%) and symbiotic corals (21.92 ± 19.46%) (Fig. 3 C). However, the relative amount of residual tissue remaining in the lesion site was greater among aposymbiotic (9.52 ± 5.33%) compared to symbiotic (7.03 ± 3.99%) colonies. As such, while there was no significant difference in final recovery between symbiotic states, the effect of residual tissue on final recovery was significantly greater in symbiotic corals (Table 1).

Mortality of the remaining colony tissue was observed to occur almost exclusively on the coenenchyme rather than the polyps themselves. Mortality was highest early in the experimental period, though early recovery was greater (Figs. 3, 4). A significant interaction between wound treatment and symbiotic state was observed ($F(3, 30) = 3.437$, $p = 0.029$, Table 3), and post hoc Tukey comparisons indicated that the relative mortality of the symbiotic corals in the 50% separate wound group (29.58 ± 26.23%) exceeded that of the symbiotic (5.68 ± 4.22%) and aposymbiotic (3.88 ± 1.57%) corals of the 25% wound group, the symbiotic corals of the 50% adjacent wound group (4.80 ± 0.69%), and the aposymbiotic corals in the 50% separate wound group (7.95 ± 4.94) (Fig. 4 A, B). It should be noted that the large variance in the mortality of symbiotic corals in

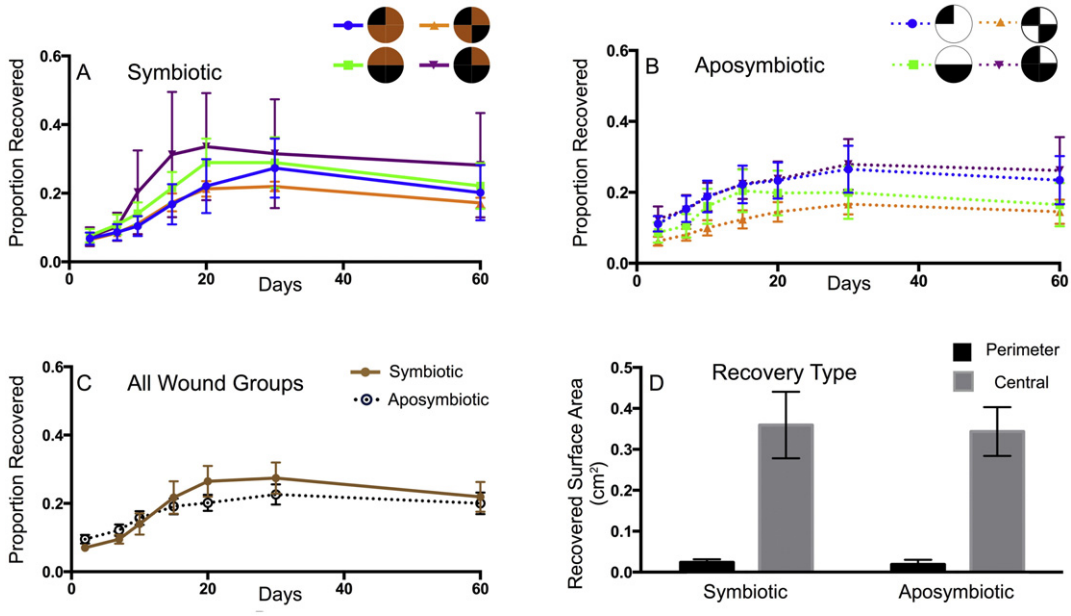

Fig. 3. Healing patterns in *A. poculata*. Mean proportion of the lesion area recovered for each wound group of (A) symbiotic *A. poculata* colonies, (B) aposymbiotic *A. poculata* colonies, and (C) all wound groups of symbiotic and aposymbiotic colonies combined. (D) mean surface area of new tissue by recovery type (central vs. perimeter). Error bars represent standard error from the mean.

the 50% separate wound group (Fig. 4 A) was driven by a single individual exhibiting relatively low mortality.

Maximum quantum yield (F_v/F_m) values were significantly greater among symbiotic *A. poculata* at every day on which measurements were taken, but temporal trends differed between symbiotic states. Patterns in photosynthetic efficiency were best explained by an interaction effect between symbiotic state and day, with coral specimen as a random effect (Table 4). There was little support ($\Delta AIC_c > 4$) for models containing only main effects of these two parameters or symbiotic state by itself, and essentially no support ($\Delta AIC_c > 10$) for all other candidate models, including those containing wound treatment and all other possible combinations of main and interaction effects (Table 2). Mean F_v/F_m values across the time series were 0.61 ± 0.06 for wounded symbiotic *A. poculata*, and 0.46 ± 0.09 for wounded aposymbiotic colonies. Wounding had no effect on F_v/F_m, and the values for the control group averaged 0.62 ± 0.07 for symbiotic corals, and 0.45 ± 0.08 for aposymbiotic corals. For symbiotic *A. poculata*, F_v/F_m values on day 3 were significantly greater than those taken prior to wounding, or 30 and 60 days after wounding (Fig. 5 C). Conversely, for aposymbiotic *A. poculata*, F_v/F_m values taken on days 3, 15, 30, and 60 were each significantly greater than those taken prior to wounding, and measurements taken on day 60 were significantly greater than those from days 3, and 30 (Fig. 5 C).

Mean baseline polyp color PC values for symbiotic and aposymbiotic *A. poculata* established prior to wounding were 2.84 ± 0.47, and 1.11 ± 0.51 respectively, indicating total approximated chlorophyll concentrations of 0.46 ± 0.14 and 0.10 ± 0.06 μg cm^{-2}. The most parsimonious model for approximated chlorophyll density included only symbiotic state as a fixed effect, with coral specimen as a random effect. Approximated chlorophyll density was significantly greater among symbiotic *A. poculata* for the duration of the experiment (Fig. 5, Table 5) There was very little support ($\Delta AIC_c > 9$) for all other candidate models considered, including those incorporating wound treatment, day, and all possible combinations of interaction and main effects (Table 2).

Dissection of preserved and decalcified *A. poculata* specimens showed unexpectedly deep polyp penetration, extending roughly 4–6 mm into the skeleton. Specimens wounded according to the same protocol as those in this study were similarly dissected, revealing preserved tissue remnants in the lesion site immediately after wounding.

Table 1
Model summary of beta regression for central recovery with logit link function. Pseudo-$R^2 = 0.555$, Log-likelihood = 39.651, $AIC_c = -67.427$. ϕ (Phi) serves as the precision parameter of the beta density for this model. Aposymbiotic *A. poculata* serve as the reference category for all effects that include symbiotic state.

Parameter	Estimate	Std. error	z	p
Intercept (aposymbiotic)	−2.519	0.324	−7.776	<0.001
Symbiotic	−0.447	0.449	−0.997	0.319
Residual tissue	10.896	2.645	4.119	<0.001
Symbiotic × residual tissue	11.272	4.445	2.536	0.011
ϕ (Phi)	15.550	3.550		

4. Discussion

This study makes several new contributions to our understanding of symbiosis in the recovery and maintenance of *A. poculata* coral hosts. First, the physical and temporal patterns of wound recovery in *A. poculata* were examined and no difference between symbiotic states in final recovery was observed, suggesting that the absence of symbiosis does not preclude timely recovery. However, the effect of residual tissue on final recovery was significantly greater in symbiotic corals, suggesting that photosynthetic capability is advantageous. Most importantly, previously described healing models ('localized regeneration (LR)', 'colony integration (CI)', or 'Phoenix effect (PE)') were evaluated and the dominance of recovery originating from within the lesion site rather than its perimeter suggests that *A. poculata* exhibited primarily PE recovery in this study. This suggests that the mechanism of wound recovery across species should be considered when using healing as a proxy for coral health, since recovery style may influence the rate and likelihood of recovery.

The prevalence of PE recovery in *A. poculata* is likely due to two factors. Firstly, the deep corallite structure provides a refuge for polyp tissue against all but the most invasive lesioning agents. The presence of tissue remnants deep within dissected corallites immediately after wounding strongly suggests that even the most severe natural wounding would not fully extirpate tissue from the lesion site, and that residual polyp tissue would be the main driver of tissue recovery in this species. Corals of the genus *Porites* exhibit deep tissue

Table 2
Model selection results for recovery (a), photosynthetic efficiency (b), and approximated chlorophyll density (c). Only candidate models within >10 ΔAIC_c are presented. For each outcome, models are ranked from most plausible ($\Delta AIC_c = 0$) to least plausible; df is the degrees of freedom for each model.

Response variable	Model parameters	Log likelihood	df	ΔAIC_c
(a) Recovery	Symbiotic, residual tissue, symbiotic × residual tissue	39.651	5	0.00
	Symbiotic, residual tissue, symbiotic × residual tissue, initial live tissue	39.844	6	2.45
	Symbiotic, residual tissue, symbiotic × residual tissue, P:SA	39.722	6	2.69
	Symbiotic, residual tissue	36.344	4	3.95
	Symbiotic, residual tissue, symbiotic × residual tissue, initial live tissue, P:SA	40.177	7	4.81
	Symbiotic, residual tissue, P:SA	36.743	5	5.82
	Symbiotic, residual tissue, initial live tissue	36.509	5	6.28
	Symbiotic, residual tissue, symbiotic × residual tissue, wound treatment	40.649	8	7.09
	Symbiotic, residual tissue, initial live tissue, P:SA	37.381	6	7.37
	Residual tissue	33.073	3	7.99
	Residual tissue, P:SA	34.070	4	8.50
	Symbiotic, residual tissue, symbiotic × residual tissue, P:SA, wound treatment	41.058	9	9.74
(b) Photosynthetic efficiency	Symbiotic, day, symbiotic × day	287.255	12	0.00
	Symbiotic, day	280.632	8	4.53
	Symbiotic	276.276	4	4.80
(c) Approximated chlorophyll density	Symbiotic	106.446	4	0.00
	Symbiotic, day	105.878	8	9.69

penetration as well (Davies, 1991), and are also one of the only other coral taxa in which PE healing has been documented (Roff et al., 2014; Jokiel et al., 1993; Meesters et al., 1992), suggesting a morphological basis for this healing mode that may be consistent across taxa. Second, *A. poculata* was observed to have a thin coenenchyme, often partially missing prior to collection or lost to varying extents during the study. This observation may indicate low levels of colony integration (Soong and Lang, 1992), potentially impeding translocation of photosynthate to the wound site from healthy polyps as required for LR and CI healing.

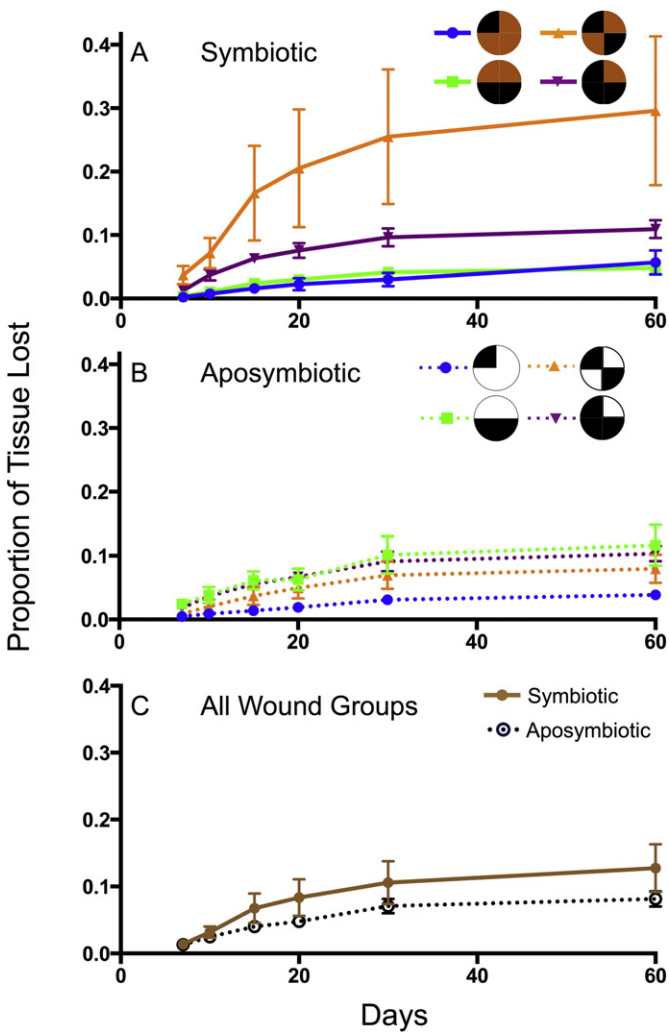

Fig. 4. Tissue mortality patterns in *A. poculata*. Mean proportion of living tissue lost for each wound group of (A) symbiotic *A. poculata* colonies, (B) aposymbiotic *A. poculata* colonies, and (C) all wound groups of symbiotic and aposymbiotic colonies combined. Error bars represent standard error from the mean.

Although LR is among the more common mechanisms of healing in corals (Meesters et al., 1994; Bak and Steward-Van Es, 1980; Meesters et al., 1997a; Van Woesik, 1998), in this study there was insubstantial evidence that either wound treatment or wound P:SA influenced recovery. This is consistent with biological expectations, as the LR hypothesis assumes that no tissue remains in the lesion site after wounding (Meesters et al., 1997a) and is predicated on the expectation that recovery originates from the lateral margins of the wound (Meesters et al., 1994; Bak and Steward-Van Es, 1980; Meesters et al., 1997a; Van Woesik, 1998). Our decalcified and dissected colonies indicated tissue remnants in the lesion site, providing further support for a PE-driven recovery mechanism. Moreover, the CI model also assumes that recovery originates from the lesion perimeter, which was seldom observed.

The amount of residual tissue remaining after wounding had a significant impact on final recovery for multiple possible reasons: (1) given equal rates of recovery, a coral with a greater amount of initial tissue has a 'head start' over other colonies. However, the regression coefficients for residual tissue indicate that its influence is beyond a mere additive contribution (Table 1). (2) If 'Phoenix effect' recovery proceeds only from residual tissue fragments, the number and size of these fragments may constitute a limiting factor in recovery, particularly if they depend on their own photosynthate for expansion. In both LR and CI, the energy for recovery is provided by translocation of photoassimilates from healthy polyps adjacent to, or distant from the lesion site respectively (Taylor, 1977; Gladfelter, 1983; Meesters et al., 1997a, 1994; Oren et al., 2001, 1997). If residual tissue fragments are isolated from the rest of the colony, they may need to provide their own energy for expansion and development of differentiated structures. Residual tissue fragments were never observed bearing feeding tentacles, but they did appear to retain the normal pigmentation of the living colony, suggesting the continued presence of *S. psygmophilum* in symbiotic colonies. If photosynthesis is indeed the primary energy source for recovery, then

Table 3
ANOVA summary for adjusted mortality.

Parameter	df	SSE	MSE	F	p
Symbiotic state	1	0.020	0.020	1.920	0.176
Wound treatment	3	0.109	0.036	3.492	0.028
Symbiotic state × wound treatment	3	0.107	0.036	3.437	0.029
Residuals	30	0.311	0.010		

Table 4
Summary of REML fitted linear mixed effects model for F_v/F_m. For symbiotic state, aposymbiotic corals serve as the reference category, and for day, the pre-wound measurements serve as the reference category. Log likelihood = 287.255, AIC_c = −549.193. Coral specimen was included in the model as a random effect.

Parameter	Estimate	Std. error	t	p
Intercept (aposymbiotic)	0.389	0.014	28.086	<0.001
Symbiotic	0.187	0.020	9.535	<0.001
Day 3	0.064	0.018	3.477	0.006
Day 15	0.079	0.018	4.326	<0.001
Day 30	0.073	0.018	3.976	0.001
Day 60	0.133	0.018	7.232	<0.001
Symbiotic × day 3	0.026	0.026	0.998	0.320
Symbiotic × day 15	−0.025	0.026	−0.966	0.335
Symbiotic × day 30	−0.051	0.026	−1.972	0.05
Symbiotic × day 60	−0.124	0.026	−4.766	<0.001

a greater area of tissue in the wound site would achieve greater net carbon fixation than a smaller one (Jokiel and Morrissey, 1986), providing more energy for expansion and differentiation. This could also explain why symbiotic *A. poculata*, with their greater symbiont/chlorophyll densities (Dimond and Carrington, 2007, 2008) and photosynthetic capacity (Jacques et al., 1983), achieved more recovery per unit area of residual tissue, even with daily ad libitum feedings to all colonies.

Previous descriptions of PE recovery have occurred in response to hyposalinity (Krupp et al., 1992; Jokiel et al., 1993) and thermally induced bleaching (Roff et al., 2014). Interestingly, both of these sources of mortality are mediated by contact of the colony surface with the surrounding water, and may not impact deeper portions of the coral. By contrast, more invasive sources of mortality such as excavating sponges (Chaves-Fonnegra and Zea, 2011), and parrotfish (Bonaldo et al., 2011; Rotjan and Lewis, 2005, 2008) severe hurricane damage (Woodley et al., 1981; Bythell et al., 1993), dynamite, and ship groundings (Riegl, 2001) may be less likely to illicit PE healing. Thus, the encouragingly rapid recovery that Roff et al. (2014) observed in *Porites* after mass bleaching may not be as likely in response to more invasive wounding agents.

Table 5
Summary of REML fitted linear mixed effects model for chlorophyll density. The reference category for symbiotic state is the aposymbiotic corals. Log likelihood = 106.45, AIC_c = −204.69. Coral specimen was included in the model as a random effect.

Parameter	Estimate	Std. error	t	p
Intercept (aposymbiotic)	0.150	0.028	5.419	<0.001
Symbiotic	0.281	0.039	7.153	<0.001

The only significant differences in relative mortality involved the symbiotic corals exposed to the 50% separate wound treatment. The wounding treatment itself likely contributed to the greater mortality of this group by disrupting the living colony's integrity more extensively than the other lesion types (Fig. 1). While this wounding pattern was designed to maximize the perimeter of living tissue in contact with the wound site to potentially provide greater opportunity for recovery, it also achieved maximal disruption of colony cohesion. Only the symbiotic colonies exposed to this wounding treatment suffered greater mortality, adding to a small but growing body of evidence that algal symbiosis can exert an energetic demand on corals (Sachs and Wilcox, 2006; Burmester and Rotjan, pers. obs.).

As anticipated, photosynthetic efficiency and approximated chlorophyll density were both greater among symbiotic compared to aposymbiotic *A. poculata* on all days measured. Neither parameter differed among wound treatments, suggesting that the greater mortality

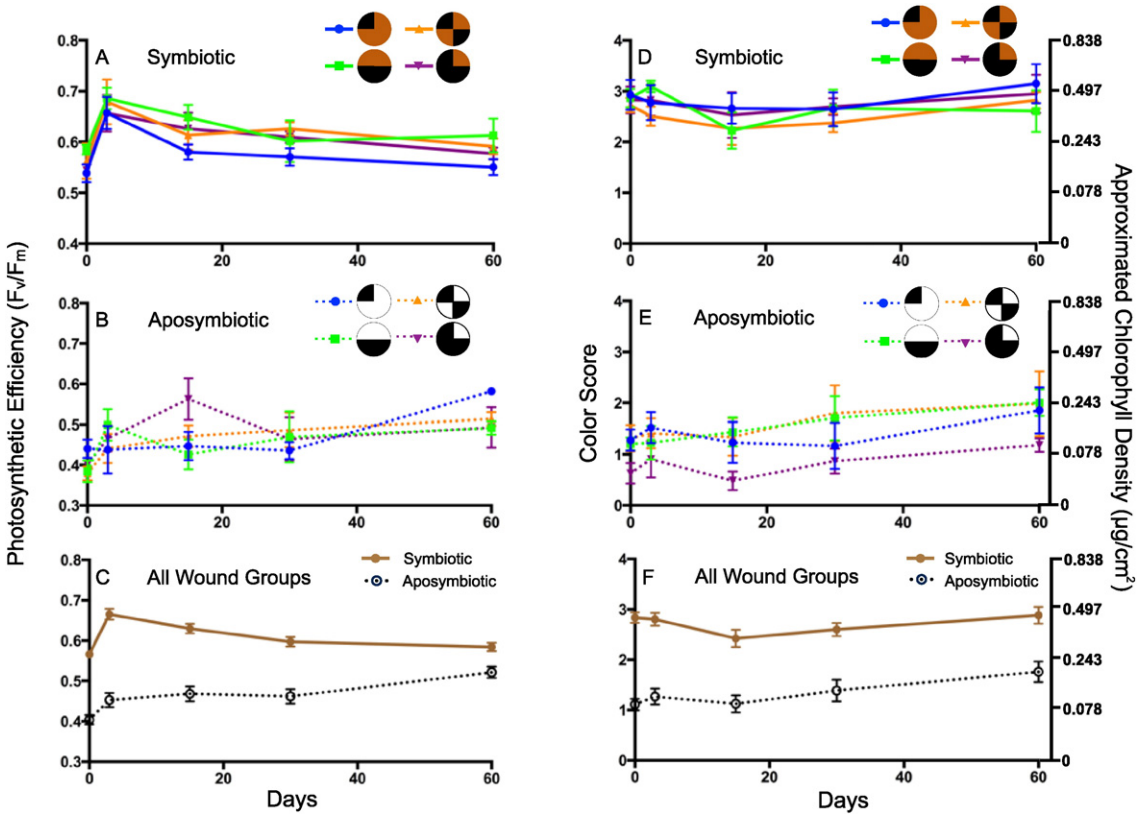

Fig. 5. Photosynthetic efficiency and chlorophyll density in *A. poculata*. Mean photosynthetic efficiency (F_v/F_m) over time for each wound group of (A) symbiotic *A. poculata* colonies, (B) aposymbiotic *A. poculata* colonies, and (C) all wound groups combined for symbiotic and aposymbiotic colonies (including unwounded controls). Mean color score and approximated chlorophyll density over time for each wound group of (D) symbiotic *A. poculata* colonies, (E) aposymbiotic *A. poculata* colonies, and F) all wound groups combined for symbiotic and aposymbiotic colonies. Error bars represent standard error from the mean.

observed among symbiotic corals of the 50% separate wound group was due to the wounding pattern itself. Moreover, the consistently greater F_v/F_m values of symbiotic *A. poculata* indicate greater photosynthetic capacity of PS II, further suggesting that elevated symbiont densities and photosynthetic yield are responsible for the greater healing capacity observed in these colonies. While chlorophyll density estimated through polyp color cannot distinguish between changes in symbiont densities versus changes in the chlorophyll densities of symbionts (Warner et al., 2002; Cummings, 1983; Fitt et al., 2000), it has nonetheless been demonstrated as a viable and non-destructive tool to estimate symbiosis in *A. poculata* (Dimond and Carrington, 2007). Because direct zooxanthellae counts are destructive, the in situ measurement of symbiont densities would have been a confounding factor, acting as a de facto wound beyond our experimental construct. Interestingly, the approximated chlorophyll densities estimated in this study for both symbiotic states were lower than those described by Dimond and Carrington (2007), possibly reflecting the effects of stress induced by wounding and/or captivity on symbiont/pigment concentrations.

5. Conclusions

Symbiotic and aposymbiotic varieties of *A. poculata* exist sympatrically, suggesting that both symbiotic states regularly persist in nature. One potential advantage conferred by symbiosis is greater healing capacity with single wounds, regardless of size, most likely attributable to increased photosynthetic yield. However, our results indicate that when multiple large wounds are inflicted, there is little impact on wound size or position on recovery, and corals in both symbiotic and aposymbiotic states were still able to demonstrate recovery.

This study supports a morphological basis for PE recovery in corals, which can be used to predict the occurrence of this healing pattern in response to lesions of varying invasiveness. Considering that PE recovery has been implicated in the rapid, large scale recovery of corals with deep-bodied polyps, these results will have useful applications towards predicting the ecological consequences of disturbances for reefs, and future studies should consider the role of polyp depth in colony recovery potential. Additionally, our results indicate that different configurations of the same relative lesion size can have markedly different impacts on colony health, informing the use of recovery from artificially inflicted wounds as a proxy for coral status.

Declared author contributions

LD: Carried out the experiment, analyzed the images, performed the statistical analyses, graphed the data, wrote the manuscript, and contributed funding.

EMB: Collected specimens, collaborated in project design and implementation, assisted with figure creation, and manuscript writing, and contributed resources and funding.

LK: Contributed useful conversations and input.

RR: Conceived and designed the experiment, helped to perform the experiment, assisted with graphics, and manuscript writing and contributed space, other resources, and funding.

All authors have agreed to the submission of the article and declare no conflicts of interest.

Role of the funding source

This research was funded by a grant from the PADI Foundation awarded to EMB, an Undergraduate Research Grant from the Boston University Marine Program awarded to LD, and a grant from the Cell Signaling Technologies Environmental Small Grants Program awarded to RR. Personnel from both Boston University and the New England Aquarium were directly involved in the study design, collection, analysis and interpretation of data, in writing the report, and in the decision to submit the article for publication. However, institutional funding sources were not involved in any of the above.

Acknowledgments

We would like to thank Tasia Blough, Katrina Malakhoff, Aaron Pilnick, Jesse Mathews, Julio Camperio-Ciani, Georgia Luddecke and Nicholas Lawrence for the assistance in data collection and aquarium husbandry. We also thank Jeff Chabot for programming RGB color analysis in MATLAB, and Thomas Buehrens for the statistical advice. [**SS**]

References

Bak, R.P.M., Steward-Van Es, Y., 1980. Regeneration of superficial damage in the scleractinian corals *Agaricia agaricites* f. *purpurea* and *Porites astreoides*. Bull. Mar. Sci. 30, 883–887.

Bonaldo, R.M., Krajewski, J.P., Bellwood, D.R., 2011. Relative impact of parrotfish grazing scars on massive *Porites* corals at Lizard Island, Great Barrier Reef. Mar. Ecol. Prog. Ser. 423, 223–233.

Brown, B.E., Ambarsari, I., Warner, M.E., Fitt, W.K., Dunne, R.P., Gibb, S.W., Cummings, D.G., 1999. Diurnal changes in photochemical efficiency and xanthophyll concentrations in shallow water reef corals: evidence for photoinhibition and photoprotection. Coral Reefs 18, 99–105.

Burnham, K.P., Anderson, D.R., 2002. Model Selection and Multimodel Inference: A Practical Information-theoretic Approach. Springer Science & Business Media, New York.

Bythell, J.C., Gladfelter, E.H., Bythell, M., 1993. Chronic and catastrophic natural mortality of three common Caribbean reef corals. Coral Reefs 12, 143–152.

Chaves-Fonnegra, A., Zea, S., 2011. Coral colonization by the encrusting excavating Caribbean sponge *Cliona delitrix*. Mar. Ecol. 32, 162–173.

Core Team, R., 2015. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria.

Crawley, M.J., 1983. Herbivory: The Dynamics of Animal–Plant Interactions. Blackwell Scientific Publications, Oxford.

Cummings, C., 1983. The Biology of *Astrangia danae* PhD Dissertation University of Rhode Island, Kingston.

Davies, P.S., 1991. Effect of daylight variations on the energy budgets of shallow-water corals. Mar. Biol. 108, 137–144.

Denis, V., Debreuil, J., De Palmas, S., Richard, J., Guillaume, M.M., Bruggemann, J.H., 2011. Lesion regeneration capacities in populations of the massive coral *Porites lutea* at Réunion Island: environmental correlates. Mar. Ecol. Prog. Ser. 428, 05–117.

Dimond, J., Carrington, E., 2007. Temporal variation in the symbiosis and growth of the temperate scleractinian coral *Astrangia poculata*. Mar. Ecol. Prog. Ser. 348, 161–172.

Dimond, J., Carrington, E., 2008. Symbiosis regulation in a facultatively symbiotic temperate coral: zooxanthellae division and expulsion. Coral Reefs 27, 601–604.

Edmunds, P.J., Gates, R.D., 2003. Has coral bleaching delayed our understanding of fundamental aspects of coral–dinoflagellate symbioses? Bioscience 53, 976–980.

Ferrari, S., Cribari-Neto, F., 2004. Beta regression for modelling rates and proportions. J. Appl. Stat. 31, 799–815.

Fisher, E.M., Fauth, J.E., Hallock-Muller, P., Woodley, C.M., 2007. Lesion regeneration rates in reef-building corals *Montastraea* spp. as indicators of colony condition. Mar. Ecol. Prog. Ser. 339, 61–71.

Fitt, W.K., McFarland, F.K., Warner, M.E., Chilcoat, G.C., 2000. Seasonal patterns of tissue biomass and densities of symbiotic dinoflagellates in reef corals and relation to coral bleaching. Limnol. Oceanogr. 45, 677–685.

Gladfelter, E.H., 1983. Circulation of fluids in the gastrovascular system of the reef coral *Acropora cervicornis*. Biol. Bull. 165, 619–636.

Glynn, P.W., Peters, E.C., Muscatine, L., 1985. Coral tissue microstructure and necrosis: relation to catastrophic coral mortality in Panama. Dis. Aquat. Org. 1, 29–37.

Hall, V.R., 2001. The response of *Acropora hyacinthus* and *Montipora tuberculosa* to three different types of colony damage: scraping injury, tissue mortality and breakage. J. Exp. Mar. Biol. Ecol. 264, 209–223.

Harmelin, J.G., Marinopoulos, J., 1994. Population structure and partial mortality of the gorgonian *Paramuricea clavata* (Risso) in the north-western Mediterranean (France, Port-Cros Island). Mar. Life 4, 5–13.

Holcomb, M., McCorkle, D.C., Cohen, A.L., 2010. Long-term effects of nutrient and CO_2 enrichment on the temperate coral *Astrangia poculata* (Ellis and Solander, 1786). J. Exp. Mar. Biol. Ecol. 386, 27–33.

Holcomb, M., Cohen, A.L., McCorkle, D.C., 2012. An investigation of the calcification response of the scleractinian coral *Astrangia poculata* to elevated pCO_2 and the effects of nutrients, zooxanthellae and gender. Biogeosciences 9, 29–39.

Jackson, J.B.C., Winston, J.E., 1981. Modular growth and longevity in bryozoans. In: Larwood, G.P., Nielson, C. (Eds.), Recent and Fossil Bryozoa. Olsen & Olsen, Fredensborg, pp. 121–126.

Jacques, T.G., Pilson, M.E.Q., 1980. Experimental ecology of the temperate scleractinian coral *Astrangia danae*. I. Partition of respiration, photosynthesis and calcification between host and symbionts. Mar. Biol. 60, 167–178.

Jacques, T.G., Marshall, N., Pilson, M.E.Q., 1983. Experimental ecology of the temperate scleractinian coral *Astrangia danae*. Mar. Biol. 76, 135–148.

Jayewardene, D., 2010. Experimental determination of the cost of lesion healing on *Porites compressa* growth. Coral Reefs 29, 131–135.

Jokiel, P.L., Morrissey, J.I., 1986. Influence of size on primary production in the reef coral *Pocillopora damicornis* and the macroalga *Acanthophora spicifera*. Mar. Biol. 91, 15–26.

Jokiel, P.L., Hunter, C.L., Taguchi, S., Watarai, L., 1993. Ecological impact of a fresh-water "reef kill" in Kaneohe Bay, Oahu, Hawaii. Coral Reefs 12, 177–184.
Kramarsky-Winter, E., Loya, Y., 1996. Regeneration versus budding in fungiid corals: a trade-off. Mar. Ecol. Prog. Ser. 134, 179–185.
Kramarsky-Winter, E., Loya, Y., 2000. Tissue regeneration in the coral *Fungia granulosa*: the effect of extrinsic and intrinsic factors. Mar. Biol. 137, 867–873.
Krupp, D.A., Jokiel, P.L., Chartrand, T.S., 1992. Asexual reproduction by the solitary Scleractinian coral *Fungia scutaria* on dead parent coralla in Kaneohe Bay, Oahu, Hawaiian Islands. In: Richmond, B.M. (Ed.), Proceedings of the Seventh International Coral Reef Symposium, Hawaii, pp. 527–534.
LaJeunesse, T.C., Parkinson, J.E., Reimer, J.D., 2012. A genetics-based description of *Symbiodinium minutum* sp. nov. and *S. psygmophilum* sp. nov. (Dinophyceae), two dinoflagellates symbiotic with cnidaria. J. Phycol. 48, 1380–1391.
Lirman, D., 2000. Lesion regeneration in the branching coral *Acropora palmata*: effects of colonization, colony size, lesion size, and lesion shape. Mar. Ecol. Prog. Ser. 197, 209–215.
Marsh Jr., J.A., 1970. Primary productivity of reef-building calcareous red algae. Ecology 51, 255–263.
Mascarelli, P.E., Bunkley-Williams, L., 1999. An experimental field evaluation of healing in damaged, unbleached and artificially bleached star coral, *Montastraea annularis*. Bull. Mar. Sci. 65, 577–586.
Meesters, E.H., Bak, R.P.M., 1994. Effect of coral bleaching on tissue regeneration potential and colony survival. Mar. Ecol. Prog. Ser. 96, 189–198.
Meesters, E.H., Bos, A., Gast, G.J., 1992. Effects of sedimentation and lesion position on coral tissue regeneration. Proceedings of the 7th International Coral Reef Symposium. Vol. 2, pp. 681–688.
Meesters, E.H., Noordeloos, M., Bak, R.P.M., 1994. Damage and regeneration links to coral growth in the reef-building coral *Montastrea annularis*. Mar. Ecol. Prog. Ser. 112, 119–128.
Meesters, E.H., Wesseling, I., Bak, R.P.M., 1996. Partial mortality in three species of reef-building corals and the relation with colony morphology. Bull. Mar. Sci. 58, 838–852.
Meesters, E.H., Pauchli, W., Bak, R.P.M., 1997a. Predicting regeneration of physical damage on a reef-building coral by regeneration capacity and lesion shape. Mar. Ecol. Prog. Ser. 146, 91–99.
Meesters, E.H., Wesseling, I., Bak, R.P.M., 1997b. Coral colony tissue damage in six species of reef-building corals: partial mortality in relation with depth and surface area. J. Sea Res. 37, 131–144.
Meszaros, A., Bigger, C., 1999. Qualitative and quantitative study of wound healing processes in the coelenterate, *Plexaurella fusifera*: spatial, temporal, and environmental (light attenuation) influences. J. Invertebr. Pathol. 73, 321–331.
Mistri, M., Ceccherelli, V.U., 1995. Damage and partial mortality in the gorgonian *Paramuricea clavata* in the Strait of Messina (Tyrrhenian Sea). Mar. Life Vol. 5, 43–49 Marseille.
Mitchelmore, C.L., Ringwood, A.H., Weis, V.M., 2003. Differential accumulation of cadmium and changes in glutathione levels as a function of symbiotic state in the sea anemone *Anthopleura elegantissima*. J. Exp. Mar. Biol. Ecol. 284, 71–85.
Oren, U., Rinkevich, B., Loya, Y., 1997. Oriented intra-colonial transport of super (14) C labeled materials during coral regeneration. Mar. Ecol. Prog. Ser. 161, 117–122.
Oren, U., Benayahu, Y., Lubinevsky, H., Loya, Y., 2001. Colony integration during regeneration in the stony coral *Favia favus*. Ecology 82, 802–813.
Peters, E.C., Pilson, M.E.Q., 1985. A comparative study of the effects of sedimentation on symbiotic and asymbiotic colonies of the coral *Astrangiadanae* Milne Edwards and Haime 1849. J. Exp. Mar. Biol. Ecol. 92, 215–230.
Peters, E.C., Cairns, S.D., Pilson, M.E.Q., Wells, J.W., Jaap, W.C., Lang, J.C., Vasleski, C.E., St Pierre-Gollahon, L., 1988. Nomenclature and biology of *Astrangia poculata* (=*A. danae*, =*A. astreiformis*) (Cnidaria: Anthozoa). Proc. Biol. Soc. Wash. 101, 234–250.
Riegl, B., 2001. Degradation of reef structure, coral and fish communities in the Red Sea by ship groundings and dynamite fisheries. Bull. Mar. Sci. 69, 595–611.
Rinkevich, B., 1996. Do reproduction and regeneration in damaged corals compete for energy allocation? Mar. Ecol. Prog. Ser. 143, 297–302.
Rinkevich, B., Loya, Y., 1989. Reproduction in regenerating colonies of the coral *Stylophora pistillata*. Environ. Qual. Ecos. Stab. 4, 257–265.
Roff, G., Bejarano, S., Bozec, Y.M., Nugues, M., Steneck, R.S., Mumby, P.J., 2014. *Porites* and the Phoenix effect: unprecedented recovery after a mass coral bleaching event at Rangiroa Atoll, French Polynesia. Mar. Biol. 161, 1385–1393.
Rotjan, R.D., Lewis, S.M., 2005. Selective predation by parrotfishes on the reef coral *Porites astreoides*. Mar. Ecol. Prog. Ser. 305, 193–201.
Rotjan, R.D., Lewis, S.M., 2008. Impact of coral predators on tropical reefs. Mar. Ecol. Prog. Ser. 367, 73–91.
Sachs, J.L., Wilcox, T.P., 2006. A shift to parasitism in the jellyfish symbiont *Symbiodinium microadriaticum*. Proc. R. Soc. Lond. B Biol. Sci. 273, 425–429.
Soong, K., Lang, J.C., 1992. Reproductive integration in reef corals. Biol. Bull. 183, 418–431.
Szmant-Froelich, A., Pilson, M.E.Q., 1980. The effects of feeding frequency and symbiosis with zooxanthellae on the biochemical composition of *Astrangia danae* Milne Edwards & Haime 1849. J. Exp. Mar. Biol. Ecol. 48, 85–97.
Szmant-Froelich, A., Pilson, M.E.Q., 1984. Effects of feeding frequency and symbiosis with zooxanthellae on nitrogen metabolism and respiration of the coral *Astrangia danae*. Mar. Biol. 81, 153–162.
Taylor, D.L., 1977. Intra-colonial transport of organic compounds and calcium in some Atlantic reef corals. International Proceedings of the 3rd International Coral Reef Symposium. Vol. 1, pp. 431–436.
Titlyanov, E.A., Titlyanova, T.V., Chapman, D., 2008. Recovery dynamics of mechanically damaged scleractinian corals of the genus *Porites* under different light intensities. Galaxea, J. Coral Reef Stud. 10, 69–76.
Van Woesik, R., 1998. Lesion healing on massive *Porites* spp. corals. Mar. Ecol. Prog. Ser. 164, 213–220.
Warner, M., Chilcoat, G., McFarland, F., Fitt, W., 2002. Seasonal fluctuations in the photosynthetic capacity of photosystem II in symbiotic dinoflagellates in the Caribbean reef-building coral *Montastraea*. Mar. Biol. 141, 31–38.
Woodley, J.D., Chornesky, E.A., Clifford, P.A., Jackson, J.B.C., Kaufman, L.S., Knowlton, N., Lang, J.C., Pearson, M.P., Porter, J.W., Rooney, M.C., Rylaarsdam, K.W., Tunnicliffe, V.J., Wahle, C.M., Wulff, J.L., Curtis, A.S.G., Dallmeyer, M.D., Jupp, B.P., Koehl, M.A.R., Neigel, J., Sides, E.M., 1981. Hurricane Allen's impact on a Jamaican coral reef. Science 214, 749–754.

Glossary

Polyp: Genetically identical functional units that make up the feeding portion of a living coral colony. Consists of a stalk terminating in an oral disk surrounded by a ring of tentacles.
Coenenchyme: The tissue layer found in between the polyps of the living coral animal.
Corallite: Walled cup-like units of the coral skeleton in which the polyps are seated.
Septa: Vertical blades arranged radially in the corallite.
Mesenteries: Thin folds of tissue found within the polyp.